

CareerVerse AI

Smart Career Guidance, Government Exam, College Finder and Placement Preparation Platform

Project Upgrade Documentation

This document explains the upgraded website plan, features, technology stack, AI usage, Google Authentication, database design, government exam guidance module, trainer module, live updates module, and implementation status.

Deliverable	Details
Frontend	React + TypeScript + Vite + Tailwind CSS
Backend blueprint	Spring Boot + MySQL + JWT + Google OAuth2 + Gemini API
Authentication	Default users removed; backend API and Google OAuth flow prepared
New major module	Government Exam Guidance, Trainer and Live Updates System

1. Project Overview

CareerVerse AI is designed as a complete guidance platform for students, parents, colleges, recruiters, mentors and admins. The upgraded version keeps the existing career guidance pages and adds government exam guidance, trainer profiles, videos, live updates, Google OAuth, database-ready login and Gemini AI API-ready tools.

- Students can explore careers, courses, colleges, entrance exams, government exams, internships and placement preparation.
- Parents can understand career paths, course cost, counselling flow and child progress.
- Recruiters and companies can later post jobs/internships and view applicants.
- Mentors and trainers can publish guidance profiles, course cards and videos.
- Admins can manage exams, official links, live updates, trainers, videos and platform content.

2. Technology Used

Layer	Technology	Purpose
Frontend	React 18 + TypeScript	Build reusable components, pages, routing and type-safe UI logic.
Build Tool	Vite	Fast local development and optimized production build.
Styling	Tailwind CSS	Clean responsive UI, cards, gradients, spacing, and mobile-first layout.
Routing	React Router	Separate routes for government exams, trainer profiles, live updates and AI advisor.
Icons	Lucide React	Modern icons for actions, cards and navigation.
Backend	Spring Boot blueprint	REST APIs, authentication, Google OAuth, Gemini AI calls and MySQL data.
Database	MySQL	Store users, exams, trainers, videos, AI history and saved exams.
Security	JWT + BCrypt	Secure sessions and hashed passwords.
AI	Gemini API	Generate exam suggestions, roadmaps, preparation plans, interview guidance and skill plans.

3. Authentication Upgrade

The earlier project used frontend demo users. In this upgrade, demo/default users are removed and the login screen now expects backend authentication.

- Removed hardcoded users from frontend.
- Removed demo credentials and fake login messages.
- Added backend API login: POST /api/auth/login.
- Added JWT token storage with local session restore.
- Added Google OAuth button for non-admin users.
- Added /auth/callback route to receive backend OAuth success response.
- Admin login is email/password only from MySQL database.

Google OAuth flow

React login page -> Spring Boot OAuth2 endpoint -> Google login -> backend checks/creates MySQL user -> backend generates JWT -> redirects to /auth/callback -> frontend stores token -> dashboard redirect.

4. API Key Setup

Google OAuth Client ID and Secret

- Open Google Cloud Console.
- Create/select a project.
- Configure OAuth consent screen.
- Create OAuth Client ID with type Web application.
- Add JavaScript origin: `http://localhost:5173`.
- Add redirect URI: `http://localhost:8080/login/oauth2/code/google`.
- Copy `GOOGLE_CLIENT_ID` and `GOOGLE_CLIENT_SECRET` to backend environment variables.

Gemini API Key

- Open Google AI Studio.
- Go to API keys.
- Create an API key for the project.
- Copy `GEMINI_API_KEY` into backend environment variables.
- Never put Gemini API key in React frontend. React should call Spring Boot, and Spring Boot should call Gemini.

Example backend environment

```
DB_URL=jdbc:mysql://localhost:3306/careerverse_ai
JWT_SECRET=your_long_secret
GOOGLE_CLIENT_ID=your_client_id
GOOGLE_CLIENT_SECRET=your_client_secret
GEMINI_API_KEY=your_gemini_key
```

5. Government Exam Guidance Module

A new full module was added to guide students through government exams, official links, preparation plans, trainer guidance, videos, vacancies and results.

Exam Category	Examples / Purpose
Civil Services	UPSC CSE - IAS, IPS, IFS, IRS and other central services.
Central Government	SSC CGL and other SSC exams for graduate-level central jobs.
Banking	IBPS PO, Clerk, SO and RRB banking recruitment.
Railway	RRB NTPC, Technician, JE and Group D exams.
Tamil Nadu State	TNPSC Group I, II, IIA, IV and other state service exams.
Teaching	TN TRB, TET, lecturer and assistant professor recruitment.
Police	TNUSRB Police Constable, SI, Jail Warder and Fireman.
NTA	CUET, NEET, JEE and other NTA entrance/recruitment exams.

Each exam detail page includes

- Exam name, category and conducting body.
- Eligibility, qualification and age limit.
- Selection process, exam pattern and syllabus focus.
- Documents required and application fee note.
- Step-by-step how to apply guide.
- Preparation roadmap and best resources.
- Official apply, notification, admit card and result links.
- Trainer cards, guidance videos and AI preparation prompt.

6. Official Link Policy

For trust and safety, government exam pages should use only official links. Admin should verify links before publishing updates. The delivered project includes official-link fields in the exam data and live-update cards.

Body	Official Website / Portal Included
UPSC	upsc.gov.in and upsconline.nic.in
SSC	ssc.gov.in
IBPS	ibps.in
RRB	rrbapply.gov.in
TNPSC	tnpsc.gov.in and apply.tnpscexams.in
TN TRB	trb.tn.gov.in
TNUSRB	tnusrb.tn.gov.in
NTA	nta.ac.in and exams.nta.ac.in

7. Trainer and Guidance Video Module

- Separate Government Exam Trainers page.
- Trainer profile cards with category, subjects, experience, language, mode, location, rating and students trained.
- Trainer detail page with about, course cards, demo video section, success highlights and guidance booking buttons.
- Guidance videos page design supports YouTube/video links, trainer name, language, category and duration.
- Admin can later add, edit and remove trainer profiles and videos from dashboard.

8. Live Updates Module

The live updates page is designed for latest notifications, vacancies, admit cards, results, answer keys, exam dates and document verification updates.

- Manual admin update method: safest for IET/project demo and avoids fake information.
- Advanced method: Spring Boot scheduler can monitor official pages and save draft update records.
- Admin verification should be required before publishing any update.
- Each update card includes exam name, update type, title, description, source, official link, status and posted date.

9. Database Design

The SQL schema file database/careerverse_schema.sql was added. It supports users, government exams, live updates, trainers, videos, AI chat history and saved exams.

Table	Purpose
users	Stores database and Google OAuth users with roles and provider type.
government_exams	Stores exam details, eligibility, syllabus and official links.
exam_updates	Stores notification, vacancy, admit card, answer key and result updates.
trainers	Stores trainer profiles, subjects, language, mode, rating and contact.
trainer_courses	Stores courses/batches offered by trainers.
guidance_videos	Stores video links for exam guidance.
ai_chat_history	Stores AI prompts and responses for logged-in users.
saved_exams	Stores user saved exams and application tracking status.

10. UI/UX Design Approach

- Clean academic and modern light theme suitable for students.
- Card-based navigation to make large content easy to scan.
- Clear CTAs such as Official Apply, View Full Guide, Find Trainers and Open AI Advisor.
- Filters and search for exam discovery and live updates.
- Role-based login screen for student, parent, college, recruiter, mentor and admin.
- Responsive layout for mobile, tablet and desktop.
- Consistent visual hierarchy with headings, badges, icons and section cards.

11. New Frontend Files Added

```
src/services/api.ts
src/pages/AuthCallback.tsx
src/data/governmentExams.ts
src/pages/government/GovernmentExams.tsx
src/pages/government/GovernmentExamDetail.tsx
src/pages/government/GovernmentExamTrainers.tsx
src/pages/government/TrainerDetail.tsx
src/pages/government/LiveGovernmentUpdates.tsx
src/pages/government/AIExamAdvisor.tsx
```

12. New Routes Added

```
/government-exams
/government-exams/:examId
/government-exam-trainers
/government-exam-trainers/:trainerId
/exam-live-updates
/ai-tools/government-exam-advisor
/auth/callback
```

13. Implementation Status

Item	Status
Demo users removed	Completed in frontend. Login now expects backend auth.
Google OAuth UI	Completed in frontend. Backend blueprint included.
JWT session storage	Completed in frontend. Backend token generation blueprint included.
Government exam pages	Completed as frontend pages with structured data.
Trainer pages	Completed as frontend pages with cards and detail page.
Live updates page	Completed as frontend design with official-link update cards.
Gemini AI advisor page	Completed with backend API call and fallback suggestion.
MySQL schema	Completed as SQL file.
Spring Boot backend	Starter blueprint included; production JWT filter still needs final implementation.
Frontend build	npm run typecheck and npm run build passed.

14. How to Run

Frontend:

```
npm install
npm run dev
npm run build
```

Backend blueprint:

```
cd backend-springboot-blueprint
copy application.properties.example to application.properties
set environment variables
mvn spring-boot:run
```

15. Final Recommendation

For IET/viva, explain that CareerVerse AI started as a frontend career guidance platform and has now been upgraded into a full-stack-ready product plan with real authentication, database design, Google OAuth, Gemini AI, government exam guidance, trainer support and live official updates. For production deployment, complete the Spring Boot JWT filter, move all static exam data into MySQL, and verify every official link through the admin dashboard before publishing.